MSD MANUAL
Professional Version

Overview of Diabetes Mellitus in Children and Adolescents

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Diabetes mellitus involves absence of insulin secretion (type 1) or peripheral insulin resistance (type 2), causing hyperglycemia. Early symptoms are related to hyperglycemia and include polydipsia, polyphagia, polyuria, and weight loss. Diagnosis is by measuring plasma glucose levels. Treatment depends on type but includes medications that reduce blood glucose levels, diet, and exercise.

(See also Diabetes Mellitus in adults.)

The types of diabetes mellitus in children are similar to those in adults, but psychosocial problems are different and can complicate treatment.

Types of Diabetes in Children and Adolescents

Type 1 diabetes is the most common type in children, accounting for two thirds of new cases in children of all racial and ethnic groups. It is one of the most common chronic childhood diseases, occurring in 1 in 300 children by age 18 (1).

Although type 1 can occur at any age, it is typically diagnosed between age 4 years and 6 years or between age 10 years and 14 years. The incidence has been increasing worldwide at a rate of 2 to 5%. Despite prior reported increases in children < age 5 years (2), this trend in this age group has not continued, and greater increases in children ages 10 to 19 years have been noted (3, 4).

Type 2 diabetes, once rare in children, has been increasing in frequency in parallel with the increase in childhood obesity (see obesity in children).

Type 2 is typically diagnosed after puberty, with the highest rate between 15 years and 19 years of age (see obesity in adolescents) (5).

Approximately 80% of children with type 2 diabetes have obesity (6). However, there is considerable heterogeneity, and the relationship between obesity and age at onset of type 2 diabetes is less clear in some ethnicities (eg, South Asian children) (7).

Monogenic forms of diabetes, previously termed maturity-onset diabetes of youth (MODY), are not considered type 1 or type 2 (although they are sometimes mistaken for them) and are uncommon (1 to 4% of cases).

Prediabetes is impaired glucose regulation resulting in intermediate glucose levels that are too high to be normal but do not meet criteria for diabetes. In adolescents with obesity, prediabetes may be transient (with reversion to normal in 2 years in 60%) or progress to diabetes, especially in adolescents who persistently gain weight.

Prediabetes is associated with the metabolic syndrome (impaired glucose regulation, dyslipidemia, hypertension, obesity).

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Etiology of Diabetes in Children and Adolescents

Most patients are categorized as having type 1 or type 2 diabetes, and this distinction is used to guide treatment. Classification is based on clinical history (age, family history, body habitus), presentation, and laboratory studies, including antibodies. However, this classification system does not fully capture the clinical heterogeneity of patients, and some patients cannot clearly be classified as having type 1 or type

2 diabetes at diagnosis. In both types 1 and 2, genetic and environmental factors can result in the progressive loss of beta-cell function that results in hyperglycemia.

Type 1 diabetes

In type 1 diabetes, the pancreas produces little to no insulin because of autoimmune destruction of pancreatic beta-cells, possibly triggered by an environmental exposure in genetically susceptible people. Inherited susceptibility to type 1 diabetes is determined by multiple genes (> 60 risk loci have been identified). Susceptibility genes are more common among some populations and explain the higher prevalence of type 1 diabetes in certain ethnic groups (eg, Scandinavians, Sardinians).

About 85% of people newly diagnosed with type 1 do not have a family history of type 1 diabetes. However, close relatives of people who have type 1 diabetes are at increased risk of diabetes (about 15 times the risk of the general population), with overall incidence 6% in siblings (> 50% in monozygotic twins) (1). The risk of diabetes for a child who has a parent with type 1 diabetes is about 3.6 to 8.5% if the father is affected and is about 1.3 to 3.6% if the mother is affected (2). Risk screening is available for relatives of people who have type 1 diabetes in an effort to identify the early stages of type 1 diabetes before symptoms occur.

Children with type 1 diabetes are at higher risk of other autoimmune disorders, particularly thyroid disease and celiac disease.

Type 2 diabetes

In type 2 diabetes, the pancreas produces insulin, but there are varying degrees of insulin resistance, and insulin secretion is inadequate to meet the increased demand caused by insulin resistance (ie, there is *relative* insulin deficiency).

Onset of type 2 diabetes often coincides with the peak of physiologic pubertal insulin resistance, which may lead to symptoms of hyperglycemia in previously compensated adolescents.

The cause of type 2 diabetes is not autoimmune destruction of beta-cells but rather a complex interaction between many genes and environmental factors, which differ among different populations and patients.

Type 2 diabetes in children is different than type 2 diabetes in adults (3). In children, decline in beta-cell function and development of diabetes-related complications are accelerated.

Risk factors for type 2 diabetes include

- Obesity
- Native American, Black, Hispanic, Asian American, and Pacific Islander heritage
- Family history (60 to 90% have a first- or second-degree relative with type 2 diabetes)
- Maternal history of type 2 diabetes or gestational diabetes during pregnancy
- Current use of atypical antipsychotic medications

Monogenic diabetes

Monogenic forms of diabetes are caused by genetic defects that are inherited in an autosomal dominant pattern, so patients typically have one or more affected family members. Unlike types 1 and 2, there is no autoimmune destruction of beta-cells or insulin resistance. Onset is usually before age 25 years.

Etiology references

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Pathophysiology of Diabetes in Children and Adolescents

In **type 1 diabetes,** lack of insulin causes hyperglycemia and impaired glucose utilization in skeletal muscle. Muscle and fat are then broken down to provide energy. Fat breakdown produces ketones, which cause acidemia and sometimes a significant, life-threatening acidosis (diabetic ketoacidosis [DKA]).

In **type 2 diabetes,** there is usually enough insulin function to prevent DKA at diagnosis, but children can sometimes present with DKA (up to 25%) or, less commonly, hyperglycemic hyperosmolar state (HHS), also referred to as hyperosmolar hyperglycemic nonketotic syndrome (HHNK), in which severe hyperosmolar dehydration occurs. HHS most often occurs during a period of stress or infection, with nonadherence to treatment regimens, or when glucose metabolism is further impaired by medications (eg, corticosteroids). Other metabolic derangements associated with insulin resistance can be present at diagnosis of type 2 diabetes and include

- Dyslipidemia (leading to atherosclerosis)
- Hypertension
- Polycystic ovary syndrome
- Obstructive sleep apnea
- Nonalcoholic steatohepatitis (fatty liver)

Atherosclerosis begins in childhood or adolescence and markedly increases risk of cardiovascular disease.

In **monogenic forms of diabetes,** the underlying defect depends on the type. The most common types are caused by defects in transcription factors that regulate pancreatic beta-cell function (eg, hepatic

nuclear factor 4-alpha [HNF-4-α], hepatic nuclear factor 1-alpha [HNF-1-α]). In these types, insulin secretion is impaired but not absent, there is no insulin resistance, and hyperglycemia worsens with age. Another type of monogenic diabetes is caused by a defect in the glucose sensor, glucokinase. With glucokinase defects, insulin secretion is normal but glucose levels are regulated at a higher set point, causing fasting hyperglycemia that worsens minimally with age.

Pearls & Pitfalls

- Despite the common misconception, diabetic ketoacidosis can occur in children with type 2 diabetes.

Symptoms and Signs of Diabetes in Children and Adolescents

In **type 1 diabetes,** initial manifestations vary from asymptomatic hyperglycemia to life-threatening DKA. Most commonly, children present with symptomatic hyperglycemia without acidosis, with several days to weeks of urinary frequency, polydipsia, and polyuria. Polyuria may manifest as nocturia, enuresis (bed-wetting), or diurnal incontinence; in children who are not toilet-trained, parents may note an increased frequency of wet or heavy diapers.

About half of children have weight loss as a result of increased catabolism and also have impaired growth.

Fatigue, weakness, candidal rashes, blurry vision (due to the hyperosmolar state of the lens and vitreous humor), and/or nausea and vomiting (due to ketonemia) may also be present initially.

In **type 2 diabetes,** the clinical presentation varies widely. Children are often asymptomatic or minimally symptomatic, and their condition may be detected only on routine testing. However, some children have a severe manifestation of symptomatic hyperglycemia, hyperglycemic hyperosmolar state or DKA.

Diagnosis of Diabetes in Children and Adolescents

- Fasting plasma glucose level ≥ 126 mg/dL (≥ 7.0 mmol/L)
- Random glucose level ≥ 200 mg/dL (≥ 11.1 mmol/L)
- Glycosylated hemoglobin (HbA1C) ≥ 6.5% (≥ 48 mmol/mol)
- Sometimes oral glucose tolerance testing
- Determination of diabetes type (eg, type 1, type 2, monogenic)

Diagnosis of diabetes in children

Diagnosis of diabetes and prediabetes is similar to that in adults, typically using fasting or random plasma glucose levels and/or HbA1C levels, and depends on the presence or absence of symptoms (see table Diagnostic Criteria for Diabetes Mellitus and Impaired Glucose Regulation).

Diabetes is diagnosed in patients *with characteristic symptoms of diabetes* and blood glucose measurements that meet either of the following criteria (1, 2):

- Random plasma glucose ≥ 200 mg/dL (≥ 11.1 mmol/L)
- Fasting plasma glucose ≥ 126 mg/dL (≥ 7.0 mmol/L); fasting is defined as no caloric intake for 8 hours

An oral glucose tolerance test is not required and should not be done if diabetes can be diagnosed by other criteria. When needed, the test should be done using 1.75 g/kg (maximum 75 g) glucose dissolved in water; a positive result is a 2-hour plasma glucose level ≥ 200 mg/dL (11.1 mmol/L). The test may be helpful in children *without symptoms* or *with mild or atypical symptoms* and may be helpful in suspected cases of type 2 or monogenic diabetes.

The HbA1C criterion is typically more useful for diagnosing type 2 diabetes, and hyperglycemia should be confirmed with a fasting or random plasma glucose. Although the HbA1C screening test is commonly used and recommended for the diagnosis of type 2 diabetes in children (3), the results of the test should be interpreted with caution in some patients. For example, in children with cystic fibrosis, HbA1C is not a recommended screening test, and the diagnosis of diabetes in these children should be based on blood glucose levels. In children with conditions causing abnormal red blood cell turnover, such as hemoglobinopathies (eg, sickle cell disease), alternative measurements (eg, fructosamine) should be considered in addition to review of blood glucose levels.

TABLE

Diagnostic Criteria for Diabetes Mellitus and Impaired Glucose Regulation

Test	Normal	Impaired Glucose Regulation	Diabetes
Fasting plasma glucose (mg/dL [mmol/L])	< 100 (< 5.6)	100–125 (5.6–6.9)	≥ 126 (≥ 7.0)
Random plasma glucose (mg/dL [mmol/L])	N/A	N/A	≥ 200 (≥ 11.1)
Oral glucose tolerance test (mg/dL [mmol/L])*	< 140 (< 7.7)	140–199 (7.7–11.0)	≥ 200 (≥ 11.1)

Test	Normal	Impaired Glucose Regulation	Diabetes
Glycosylated hemoglobin (HbA1C) (% [mmol/mol])	< 5.7 (< 39)	5.7–6.4 (39–46)	≥ 6.5 (≥ 48)

* Two-hour glucose level.

Initial evaluation

For patients who are suspected of having diabetes but who do not appear ill, initial testing to establish the diagnosis should include a basic metabolic panel, including electrolytes and glucose, and urinalysis.

For patients who are suspected of having diabetes and who are ill, testing also includes a venous or arterial blood gas, liver tests, and calcium, magnesium, phosphorus, and hematocrit levels.

Evaluation for diabetes type and stage

Additional tests should be done to differentiate between types 1 and 2 diabetes (or other types), including

- C-peptide and insulin (if not yet treated with insulin) levels
- Tests for autoantibodies against pancreatic islet cell proteins

Autoantibodies include glutamic acid decarboxylase, insulin, insulinoma-associated protein, and zinc transporter ZnT8. More than 90% of patients with newly diagnosed type 1 diabetes have ≥ 1 of these autoantibodies, whereas the absence of antibodies strongly suggests type 2 diabetes. However, about 10 to 20% of children with the type 2 diabetes phenotype have autoantibodies and are reclassified as type 1 diabetes, because such children are more likely to have a rapid progression to insulin therapy (4) and are at greater risk of developing other autoimmune disorders (4, 5, 6).

Type 1 diabetes progresses in distinct stages that are characterized by the presence of ≥ 2 islet autoantibodies (see table Type 1 Diabetes Stages). Stage is associated with risk of disease progression. For example, risk of progression to stage 3 by stage at diagnosis includes stage 1 (44% 5-year risk and an 80 to 90% 15-year risk) and stage 2 (75% 5-year risk and a 100% lifetime risk) (7). By contrast, children with a single islet autoantibody have 15% risk of progression within 10 years (8).

TABLE

Type 1 Diabetes Stages

Stages
1 positive islet autoantibody*, normal glucose levels (formerly Stage 0)

Stages
Stage 1: ≥ 2 islet autoantibodies*, presymptomatic, normal FPG levels (< 100 mg/dL [< 5.6 mmol/L]) and glucose tolerance (2-hour PG < 140 mg/dL [< 7.8 mmol/L])
Stage 2: ≥ 2 islet autoantibodies, usually presymptomatic, not meeting American Diabetes Association diagnostic criteria but with elevated FPG, HbA1C 5.7–6.4% (39–48 mmol/mol), or ≥ 10% change in HbA1C, or impaired glucose tolerance (2-hour PG 140–199 mg/dL [7.8–11.1 mmol/L] or 30-, 60-, or 90-minute PG > 200 mg/dL [11.1 mmol/L]) Stage 2a: FPG 100–115 mg/dL (5.6–6.4 mmol/L) Stage 2b: FPG 116–125 mg/dL (6.5-6.9 mmol/L)
Stage 3 (clinical type 1 diabetes): Elevated glucose or abnormal glucose tolerance meeting American Diabetes Association criteria, ≥ 2 islet autoantibodies: • Glycosylated hemoglobin (HbA1C) ≥ 6.5% (≥ 48 mmol/mol) • FPG level ≥ 126 mg/dL (≥ 7.0 mmol/L) • 2-hour OGTT ≥ 200 mg/dL (≥ 11.1 mmol/L)† • Random glucose ≥ 200 mg/dL (≥ 11.1 mmol/L) with symptoms of hyperglycemic or hypoglycemic crisis Stage 3a: Asymptomatic (requires confirmatory diagnostic testing) Stage 3b: Symptomatic
Stage 4: Long-term type 1 diabetes

* Measured in ≥ 2 samples.

† An oral glucose tolerance test is not required and should not be performed if diabetes can be diagnosed by other criteria. When needed, the test should be done using 1.75 g/kg (maximum 75 g) glucose dissolved in water; a positive result is a 2-hour plasma glucose level ≥ 200 mg/dL (11.1 mmol/L).

FPG = fasting plasma glucose; HbA1C = glycosylated hemoglobin; OGTT = oral glucose tolerence test; PG = postprandial glucose.

Data from Haller MJ, Bell KJ, Besser REJ, et al. ISPAD Clinical Practice Consensus Guidelines 2024: Screening, Staging, and Strategies to Preserve Beta-Cell Function in Children and Adolescents with Type 1 Diabetes. *Horm Res Paediatr.* 2024;97(6):529-545. doi:10.1159/000543035

Monogenic diabetes is important to recognize because treatment differs from type 1 and type 2 diabetes. The diagnosis should be considered in children with a strong family history of diabetes but who lack typical features of type 2 diabetes; that is, they have only mild fasting (100 to 150 mg/dL [5.55 to 8.32 mmol/L]) or postprandial hyperglycemia, are young and do not have obesity, and have no autoantibodies or signs of insulin resistance (eg, acanthosis nigricans). Genetic testing is available to confirm monogenic diabetes. This testing is important because some types of monogenic diabetes can progress with age.

Testing for complications

Patients with **type 2 diabetes** should have liver function tests, fasting lipid profile, and urine microalbumin:creatinine ratio done at the time of diagnosis because such children (unlike those with type 1 diabetes, in whom complications develop over many years) often have comorbidities at diagnosis, such as fatty liver, hyperlipidemia, and hypertension. Children with clinical findings suggestive of complications should also be tested:

- Obesity: Test for nonalcoholic steatohepatitis
- Daytime somnolence or snoring while sleeping: Test for obstructive sleep apnea
- Hirsutism, acne, or menstrual irregularities: Test for polycystic ovary syndrome

Testing for autoimmune diseases

Patients with **type 1 diabetes** should be tested at or near the time of diagnosis for other autoimmune diseases by measuring celiac disease antibodies and thyroid-stimulating hormone, thyroxine, and thyroid antibodies.

Testing for thyroid disease (if thyroid antibodies are negative) and celiac disease should occur every 1 to 2 years thereafter. Testing for thyroid disease should be more frequent if symptoms develop or if thyroid antibodies are positive.

Other autoimmune disorders, such as primary adrenal insufficiency (Addison disease), rheumatologic disease (eg, juvenile idiopathic arthritis, systemic lupus erythematosus, psoriasis), other gastrointestinal disorders (eg, inflammatory bowel disease, autoimmune hepatitis), and skin disease (eg, vitiligo), may also occur in children with type 1 diabetes but do not require routine screening (9).

Diagnosis references

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Treatment of Diabetes in Children and Adolescents

- Healthy food choices and exercise
- For type 1 diabetes, insulin
- For type 2 diabetes, metformin and sometimes insulin or GLP-1 agonists

Intensive education and treatment in childhood and adolescence may help achieve treatment goals, which are to normalize blood glucose levels while minimizing the number of hypoglycemic episodes and to prevent or delay the onset and progression of complications.

Lifestyle modifications that benefit all patients include

- Eating regularly and in consistent amounts
- Limiting intake of refined carbohydrates and saturated fats
- Increasing physical activity

In general, the term *diet* should be avoided in favor of *meal plan* or *healthy food choices*. The main focus is on encouraging children to eat heart-healthy meals that are low in cholesterol and saturated fats and that are suitable for all young people and their families. The goal is to improve diabetes outcomes and reduce cardiovascular risk. Clinicians should work with children with diabetes and their caregivers to create an individualized meal plan (1). To improve glycemic outcomes, patients treated with insulin should be taught to how to make prandial insulin adjustments. Setting up routines at mealtimes is also important to achieve glycemic targets.

In spite of advances in diabetes technology that have improved quality of care and glycemic control, not all patients have benefited. In the United States, children who are White or non-Hispanic have a lower rate of complications and adverse outcomes caused by poor glycemic control. Race, ethnicity, and social

determinants of health (eg, socioeconomic status, neighborhood and physical environment, food environment, health care access, social context) are associated with the ability to maintain optimal glycemic control in children with diabetes (2, 3).

Methods for monitoring glycemic control

Routine monitoring involves 1 or more of the following:

- Multiple daily glucose checks by fingerstick
- Continuous glucose monitoring
- HbA1C measurements every 3 months

Self-monitoring of blood glucose

Self-monitoring of blood glucose involves intermittent fingersticks to test capillary blood glucose using a glucose monitor (glucometer).

Self-monitoring is the traditional approach. Glucose levels are checked before all meals, before a bedtime snack, and if children have symptoms of hypoglycemia. Levels also should be checked during the night (around 2 to 3 AM) if nocturnal hypoglycemia is a concern (eg, because of hypoglycemia or vigorous exercise during the day, or when an insulin dose is increased).

Temporary adjustments are made if changes in glucose regulation are anticipated because of exercise or illness. Because exercise can lower glucose levels for up to 24 hours after activity, levels should be checked more frequently on days when children exercise or are more active. To prevent hypoglycemia, children may increase carbohydrate intake or lower insulin dosing when they anticipate increased activity. Sick-day management (measuring ketones and giving additional fluid and insulin if needed) should be used with hyperglycemia or illness.

Parents should use a journal, app, spreadsheet, smart meter, or cloud-based program to keep detailed daily records of all factors that can affect glycemic control, including blood glucose levels, timing and amount of insulin doses, carbohydrate intake, physical activity, and any other relevant factors (eg, illness, late snack, missed insulin dose).

Continuous glucose monitoring systems

Continuous glucose monitoring (CGM) systems are a common method of monitoring blood glucose levels and can replace routine self-monitoring of blood glucose for some patients. These systems are increasingly being used in all children, with the highest rates in children < 6 years old.

CGM systems are a more sophisticated and effective approach to monitoring and use a subcutaneous sensor to measure interstitial fluid glucose levels every 1 to 5 minutes and then translate the measurements into blood glucose values, thus more closely detecting glucose fluctuations that can then be acted on in real time. They transmit results wirelessly to a monitoring and display device that may be built into an insulin pump or may be a stand-alone device. By identifying times of consistent hyperglycemia and times of increased risk of hypoglycemia, CGM systems can help patients with type 1 diabetes more safely reach glycemic goals.

Given the significant burdens of monitoring requirements, CGM should be offered if available and if the patient and/or family can use the device safely. Most CGM devices now give real-time feedback about current glucose readings and trends with alarms for high and low thresholds and can replace self-monitoring of blood glucose. Compared to intermittent fingerstick monitoring, CGM systems can help lower HbA1C levels, increase the percentage of time-in-range, and lower the risk of hypoglycemia (4).

Children using a CGM device need to be able to measure blood glucose by fingerstick to calibrate their monitor and/or to verify readings if they are discordant from symptoms, but, after a brief warm-up period (1 to 2 hours), newer systems do not require regular calibration with fingerstick.

Two types of CGM systems are currently available for daily home use: real-time CGM and intermittently scanned CGM.

Real-time CGM can be used in children ≥ 2 years of age. The system automatically transmits a continuous stream of glucose data to the user in real time, provides alerts and active alarms, and also transmits glucose data to a receiver, smartwatch, or smartphone. Real-time CGM should be used as close to daily as possible for maximal benefit.

Intermittently scanned CGM can be used in children ≥ 4 years of age. It provides the same type of glucose data as real-time CGM but requires the user to purposely scan the sensor with a reader or enabled smartphone to obtain information. Similar to real-time CGM, glucose data can be transferred remotely for review by parents or health care professionals. Newer intermittently scanned CGM systems have optional alerts and alarms. Intermittently scanned CGM should be used frequently, a minimum of once every 8 hours. Children who use a CGM device need to be able to measure blood glucose with a fingerstick to calibrate their monitor and to verify glucose readings if they do not match their symptoms.

Although CGM devices can be used with any regimen, they are typically worn by insulin pump users. When used in conjunction with an insulin **pump**, the combination is known as **sensor-augmented pump therapy**. This therapy requires manual adjustment of insulin doses based on CGM results.

Other CGM systems are integrated with a pump and can also suspend the basal rate for up to 2 hours when glucose levels drop below a set threshold (low glucose suspend system) or when they are predicted to drop below a set threshold (predictive low glucose suspend system). This integration can reduce the number of hypoglycemic events, even when compared to sensor-augmented pump therapy.

Closed-loop insulin pumps can be used in children ≥ 2 years of age. These hybrid closed-loop systems automate blood glucose management through sophisticated computer algorithms that are on a smartphone or similar device and link a CGM sensor to an insulin pump to determine blood glucose levels and control insulin delivery. Delivery is controlled by suspending, increasing, or decreasing basal insulin in response to CGM values. Newer hybrid closed-loop systems allow for greater automation but do not require input for mealtime boluses by the user. These systems help more tightly control insulin dosing, limit hyperglycemic and hypoglycemic episodes, and have optional settings for sleep and exercise. A fully automated closed-loop system, sometimes known as a bihormonal (insulin and glucagon) artificial pancreas, continues to be evaluated but is not commercially available.

Type 1 diabetes management

Meal plan and exercise

In type 1 diabetes, the popularity of basal–bolus regimens and the use of carbohydrate counting (patients or caregivers estimate the amount of carbohydrate in an upcoming meal and use that amount to calculate the preprandial insulin dose) has changed meal plan strategies. In this flexible approach, food intake is not rigidly specified. Instead, meal plans are based on the child's usual eating patterns rather than on a theoretically optimal diet to which the child is unlikely to adhere, and insulin dose is matched to actual carbohydrate intake. The insulin:carbohydrate ratio is individualized but varies with age, activity level, pubertal status, and length of time from initial diagnosis. Technologic advances have allowed for greater precision and customization of insulin doses. The "500 rule" (500 divided by the total daily dose of rapid-acting insulin) can be used to calculate the initial insulin:carbohydrate ratio dose.

Insulin regimens

Insulin is the cornerstone of management of type 1 diabetes. Available insulin formulations are similar to those used in adults (see table Onset, Peak, and Duration of Action of Human Insulin Preparations). Insulin should be given before a meal, except in young children whose consumption at any given meal is difficult to predict.

Dosing requirements vary by age, activity level, pubertal status, and length of time from initial diagnosis. Within a few weeks of initial diagnosis, many patients have a temporary decrease in their insulin requirements because of residual beta-cell function (honeymoon phase). This honeymoon phase can last from a few months up to 2 years, after which insulin requirements typically range from 0.7 to 1 unit/kg/day. During puberty, patients require higher doses (up to 1.5 units/kg/day) to counteract insulin resistance caused by increased pubertal hormone levels.

Types of insulin regimens include

- Multiple daily injections (MDI) regimen using basal-bolus regimen
- Insulin pump therapy
- Fixed forms of MDI regimen or premixed insulin regimen (less common)

Most patients with type 1 diabetes should be treated with MDI regimens (multiple injections per day of basal and prandial insulin) or with insulin pump therapy as part of intensive insulin regimens with the goal of improving metabolic control.

A **basal-bolus regimen** is typically the preferred MDI regimen. In this regimen, children are given a daily baseline dose of insulin that is then supplemented by doses of short-acting insulin before each meal based on anticipated carbohydrate intake and on measured glucose levels. The basal dose can be given as a once-a-day injection (sometimes every 12 hours for younger children) of a long-acting insulin (glargine, detemir, or degludec), with supplemental boluses given as separate injections of rapid-acting insulin (usually aspart or lispro). Glargine, degludec, or detemir injections are typically given at dinner or bedtime and must not be mixed with short-acting insulin.

In **insulin pump therapy,** the basal insulin is delivered at a fixed or variable rate by a continuous subcutaneous infusion of rapid-acting insulin (CSII) through a catheter placed under the skin. Mealtime and correction boluses also are delivered via the insulin pump. The basal dose helps keep blood glucose levels in range between meals and at night. Using an insulin pump to deliver the basal dose allows for maximal flexibility; the pump can be programmed to give different rates at different times throughout the day and night.

Insulin pump therapy is increasingly being used in children because of the potential benefits of glycemic control, safety, and patient satisfaction compared to MDI regimens. This therapy is typically preferred for younger children (toddlers, preschoolers) and overall offers an added degree of control to many children (5). Others find wearing the pump inconvenient or develop sores or infections at the catheter site. Children must rotate their injection and pump sites to avoid developing lipohypertrophy. Lipohypertrophy is an accumulation of lumps of fatty tissue under the skin. The lumps occur at insulin injection sites that have been overused and can cause variation in blood glucose levels because they can prevent insulin from being absorbed consistently.

Fixed forms of MDI regimens are less commonly used. They can be considered if a basal-bolus regimen is not an option (eg, because the family needs a simpler regimen, the child or caregivers have a needle phobia, lunchtime injections cannot be given at school or daycare). In this regimen, children usually receive neutral protamine Hagedorn (NPH) insulin before eating breakfast and dinner and at bedtime and receive rapid-acting insulin before eating breakfast and dinner. Because NPH and rapid-acting insulin can be mixed, this regimen provides fewer injections than the basal-bolus regimen. However, this regimen provides less flexibility, requires a set daily schedule for meals and snack times, and has been largely supplanted by the analog insulins glargine and detemir because of the lower risk of hypoglycemia and greater flexibility.

Premixed insulin regimens use preparations of 70/30 (70% insulin aspart protamine/30% regular insulin) or 75/25 (75% insulin lispro protamine/25% insulin lispro). Premixed regimens are not a good choice but are simpler and may improve adherence because they require fewer injections. Children are given set doses twice daily, with two thirds of the total daily dose given at breakfast and one third at dinner. However, premixed regimens provide much less flexibility with respect to timing and amount of meals and are less precise than other regimens because of the fixed ratios.

Clinicians should use the most intensive management program children and their family can adhere to in order to maximize glycemic control and thus reduce the risk of long-term vascular complications.

Glycemic control and HbA1C target levels

In type 1 diabetes, blood glucose levels should be monitored by self-monitoring using fingersticks and a glucose meter or by using a CGM system to optimize control (6).

Plasma glucose targets are established to balance the need to normalize glucose levels with the risk of hypoglycemia. Typical targets for blood glucose levels are 70 to 180 mg/dL (4 to 10 mmol/L), which are in alignment with continuous glucose monitoring (CGM) targets and with greater emphasis on maintaining narrower fasting glucose levels of 70 to 110 mg/dL (4 to 8 mmol/L) (7). Treatment goals

should be individualized based on patient age, diabetes duration, access to diabetes technology (eg, insulin pumps, CGMs), comorbid conditions, and psychosocial circumstances.

HbA1C target levels for type 1 diabetes in children and adolescents have been lowered over time in an effort to reduce complications—lower HbA1C levels during adolescence and young adulthood are associated with a lower risk of vascular complications. An HbA1C target level of < 7% (< 53 mmol/mol) is appropriate for most children, but many children and adolescents do not meet this target. HbA1C levels should be measured every 3 months in all children with type 1 diabetes.

The risk of hypoglycemia in children who have hypoglycemia unawareness or lack the maturity to recognize the symptoms of hypoglycemia can limit aggressive attempts to achieve treatment goals. A less stringent HbA1C target level (< 7.5% [< 58 mmol/mol]) should be considered for such patients, whereas a more stringent target level (< 6.5% [< 48 mmol/mol]) should be reserved for the honeymoon phase (residual beta-cell function) or for select patients in whom the target can be achieved without significant hypoglycemia and without negative impacts on well-being.

An increased frequency of self-monitoring of blood glucose levels (up to 6 to 10 times per day) (6) or use of a CGM system can improve HbA1C levels because patients are better able to adjust insulin for meals, have an improved ability to correct hyperglycemic values, and are potentially able to detect hypoglycemia earlier, which prevents overcorrection (ie, excessive carbohydrate intake as treatment for hypoglycemia, resulting in hyperglycemia).

HbA1C levels correlate well to the percentage of time that blood glucose levels remain in the normal range (70 to 180 mg/dL [4 to 10 mmol/L]), termed the *percentage time-in-range*. Time-in-range is commonly used as a therapeutic goal to assess the efficacy of the insulin regimen in combination with HbA1C level. A 10% change in time-in-range corresponds to about a 0.8 percentage point change in HbA1C. For example, a time-in-range of 80% corresponds to an HbA1C level of 5.9% (41 mmol/mol), 70% corresponds to 6.7% (50 mmol/mol), 60% corresponds to 7.5% (58 mmol/mol), and 40% corresponds to 9% (75 mmol/mol) (8).

In addition to time-in-range, CGM provides information related to average sensor glucose, time-above-range (> 180 mg/dL [> 10 mmol/L]) and time-below-range (< 70 mg/dL [< 4 mmol/L]), glycemic variability, glucose management indicator, and information related to adherence (eg, active CGM time, days worn).

CGM metrics derived from use over the most recent 14 days are recommended to be used in conjunction with HbA1C level. CGM data can be reported in a standardized format. The ambulatory glucose profile (AGP) is a standardized report of the mean glucose, time-in-range, and time-below-range. When using the AGP to monitor glycemia, a goal of time-in-range of > 70% with a time-below-range of < 4% may be used as a glycemic control goal, along with the goal of an HbA1C target of < 7% (< 53 mmol/mol). Ideally, metrics recorded over a 14-day period should include (7, 9)

- Time-in-range: > 70% between 70 and 180 mg/dL (4 and 10 mmol/L)
- Time-below-range: < 4% < 70 mg/dL (< 4 mmol/L) and < 1% < 50 mg/dL (< 3 mmol/L)
- Time-above-range: < 25% > 180 mg/dL (> 10 mmol/L) and < 5% > 250 mg/dL (> 13.9 mmol/L)

Another type of CGM report is the glucose management indicator, which provides an estimated HbA1C from mean CGM glucose levels, preferably from ≥ 14 days of data.

Management of complications

Hypoglycemia is a critical but common complication in children treated with an intensive insulin regimen. Most children have several mild hypoglycemic events per week and manage them by self-treating with 15 g of fast-acting carbohydrates (eg, 4 oz of juice, glucose tablets, hard candies, graham crackers, or glucose gel).

Severe hypoglycemia, defined as an episode requiring the assistance of another person to give carbohydrates or glucagon, occurs in about 30% of children each year, and most will have had such an episode by age 18. Oral carbohydrates may be tried, but glucagon 1 mg IM is usually used if neuroglycopenic symptoms (eg, behavioral changes, confusion, difficulty thinking) prevent eating or drinking. If untreated, severe hypoglycemia can cause seizures or even coma or death. Real-time CGM devices can help children who have hypoglycemia unawareness because an alarm sounds when glucose is below a specified range or when glucose declines at a rapid rate (see Methods for monitoring glycemic control).

Ketonuria/ketonemia is most often caused by intercurrent illness but also can result from not taking enough insulin or from missing doses and can be a warning of impending DKA. Because early detection of ketones is crucial to prevent progression to DKA and minimize need for emergency department or hospital admission, children and families should be taught to check for ketones in the urine or capillary blood using ketone test strips. Blood ketone testing may be preferred in younger children, those with recurrent DKA, and insulin pump users or if a urine sample is difficult to obtain.

Ketone testing should be done whenever the child become ill (regardless of the blood sugar level) or when the blood sugar is high (typically > 240 mg/dL [13.3 mmol/L]). The presence of moderate or large urine ketone levels or blood ketone levels > 1.5 mmol/L can suggest DKA (DKA is more likely if ketone levels are > 3 mmol/L), especially if children also have abdominal pain, vomiting, drowsiness, or rapid breathing. Low urine ketone levels or blood ketone levels of 0.6 to 1.5 mmol/L must also be monitored.

When ketones are present, children are given additional short-acting insulin, typically 10 to 20% of the total daily dose, every 2 to 3 hours until ketones are cleared. Also, additional fluid should be given to prevent dehydration. This program of measuring ketones and giving additional fluid and insulin during illness and/or hyperglycemia is called **sick-day management**. Parents should be instructed to call a health care professional or go to the emergency department if ketones increase or do not clear after 4 to 6 hours, or if the clinical status worsens (eg, respiratory distress, continued vomiting, change in mental status).

Type 1 diabetes prevention

Given the high rate of progression to symptomatic stages of type 1 diabetes and the prolonged preclinical period, disease-modifying therapies have been studied in an effort to prevent or delay the onset of clinical type 1 diabetes (stage 3).

One such therapy is teplizumab. Teplizumab is an anti-CD3 monoclonal antibody. It can delay the onset of type 1 diabetes in people ≥ 8 years of age with preclinical (stage 2) diabetes. This medication is given as a single 14-day course of daily IV infusions. Adverse effects can include cytokine release syndrome (during the first 5 days), lymphopenia, rash, headache, fever, and nausea.

In a randomized, controlled study, the median time to diagnosis of stage 3 type 1 diabetes was 48 months in the teplizumab group compared to 24 months in the placebo group (10). In an extended follow-up study (median 923 days) after teplizumab treatment, the median time to diagnosis was 59.6 months for people who received teplizumab compared to 27.1 months for people who received placebo. Additionally, 50% of the people who received teplizumab did not develop type 1 diabetes compared to 22% of the people who received placebo (11).

Type 2 diabetes management

As in type 1 diabetes, lifestyle modifications, with improved nutrition and increased physical activity, are important for the management of type 2 diabetes.

Meal plan and exercise

In type 2 diabetes, most patients should be encouraged to lose weight and thus increase insulin sensitivity. For children age 3 to 13 years, a useful formula to determine the amount of calories needed is: 1000 calories + (100 × child's age in years).

Steps to improve diet and to manage caloric intake and physical activity include

- Eliminate sugar-containing drinks and foods made of refined, simple sugars (eg, processed candies, high fructose corn syrups).
- Discourage skipping meals and encourage eating meals on a schedule (preferably as a family, if possible, and without distractions from other activities, eg, television, computer, or video games).
- Avoid grazing on food throughout the day.
- Control portion size.
- Limit high-fat, high-calorie foods in the home.
- Increase fiber intake by eating more fruits and vegetables.
- Increase physical activity to 60 minutes of moderate to vigorous physical activity at least 3 days per week (preferably 5 to 7 days per week).
- Limit screen time to < 2 hours a day, including television, noneducational computer time, cell phones and other handheld devices, and video games.

Pharmacologic treatment

Insulin is started in children who present with more severe type 1 diabetes (HbA1C > 8.5% [> 69 mmol/mol] or with DKA); glargine, detemir, or premixed insulin can be used.

If acidosis is not present, metformin is usually started at the same time.

Insulin requirements may decline rapidly during the initial weeks of treatment as endogenous insulin secretion increases; insulin often can be stopped several weeks after regaining acceptable metabolic control.

Metformin is an insulin sensitizer and is the most common first-line oral antihyperglycemic medication given to patients < 18 years of age. Metformin is started as monotherapy when the initial HbA1C level is < 8.5% (< 69 mmol/mol) without acidosis or ketosis and is used in conjunction with nonpharmacologic therapy.

Metformin should be started at a low dose and taken with food to prevent nausea and abdominal pain. The dose is increased stepwise to the maximal target dose over a period of 3 to 6 weeks. If available, extended-release forms of metformin may lessen gastrointestinal adverse effects in some patients who do not tolerate standard formulations of this medication.

The goal of treatment is an HbA1C level at least < 7% (< 53 mmol/mol) and preferably < 6.5% (< 48 mmol/mol). If this cannot be achieved with metformin alone, basal insulin or liraglutide should be started. Unfortunately, about half of adolescents with type 2 diabetes ultimately fail metformin monotherapy and require insulin.

If patients do not meet targets using dual therapy with metformin and basal insulin, GLP-1 receptor agonists may be added as part of intensification therapy. Oral medications also may be considered (see below), and, in some patients, rapid-acting prandial insulin also may be needed.

Liraglutide, extended-release exenatide, and dulaglutide are glucagon-like peptide 1 [GLP-1] receptor agonists that can be used in children > 10 years of age with type 2 diabetes and can help reduce HbA1C levels. Semaglutide is another GLP-1 agonist that can be used for management of type 2 diabetes in adults and also for treatment of obesity in people > 12 years of age. These injectable noninsulin antihyperglycemic medications enhance glucose-dependent insulin secretion and slow gastric emptying.

Liraglutide is given as a daily injection, whereas extended-release exenatide, dulaglutide, and semaglutide are given as weekly subcutaneous injections, which may improve patient adherence. All of these medications promote weight loss, likely through the effect of delayed gastric emptying and appetite reduction. They are titrated to treatment doses over a period of weeks to minimize the common gastrointestinal adverse effects, especially nausea and vomiting. GLP-1 agonists can be used if metformin is not tolerated or added on if HbA1C target levels are not achieved with metformin alone within 3 months. GLP-1 agonists may be used before the initiation of insulin because they promote weight loss as well as glycemic control.

Empagliflozin, a sodium-glucose cotransporter-2 (SGLT2) inhibitor, can be used for children > 10 years of age with type 2 diabetes. SGLT2 is a glucose transporter found in the proximal tubule of the kidneys. It is responsible for approximately 90% of filtered glucose reabsorption. SGLT2 inhibitors work by blocking the coupled reabsorption of sodium and glucose from the proximal tubules, leading to increased renal excretion of glucose and lower blood glucose levels in people with type 2 diabetes. These medications are contraindicated in patients with end-stage renal disease or who are on dialysis. They can increase the risk of DKA, in some cases causing normal blood glucose levels because of

increased renal excretion of glucose. Adverse effects of these medications include increased incidence of urinary tract and genital yeast infections.

Glycemic control and HbA1C target levels

Similar to type 1 diabetes, **target fasting glucose levels** in type 2 diabetes are < 130 mg/dL (7.2 mmol/L).

Patients with type 2 diabetes usually self-monitor blood glucose levels less frequently than patients with type 1 diabetes, but frequency varies depending on the type of therapy used, fasting and postprandial glucose levels, degree of glycemic control deemed achievable, and the available resources.

Children and adolescents taking multiple daily insulin injections, those who are ill, and those with suboptimal control should monitor glucose levels at least 3 times a day (12). Those who are on stable regimens of metformin and only long-acting insulin, who are meeting their targets without hypoglycemia, can monitor less frequently, typically 2 times a day (fasting and 2 hours postprandial). The frequency of monitoring should increase if glycemic control targets are not being met, during illness, or when symptoms of hypoglycemia or hyperglycemia develop. Children and adolescents with type 2 diabetes on insulin regimens with multiple daily injections or insulin pumps sometimes use CGM systems similar to systems used by those with type 1 diabetes (6).

HbA1C target levels for type 2 diabetes in children and adolescents are similar to targets in type 1 diabetes (< 7% [< 53 mmol/mol]).

HbA1C levels should be measured every 3 months in most children with type 2 diabetes, especially if insulin is being used or metabolic control is suboptimal. Otherwise, in children with stable glucose levels, levels can be measured twice a year, although every 3 months is optimal.

More stringent targets for HbA1C (< 6.5% [< 48 mmol/mol]) and fasting blood glucose (< 110 mg/dL [6.1 mmol/L]) may be considered in patients with shorter duration of diabetes and in those treated with lifestyle interventions or metformin alone who achieve significant weight reduction.

Children with type 2 diabetes who do not meet HbA1C and/or fasting glucose targets are candidates for intensified therapy (eg, with insulin, glucagon-like peptide 1 [GLP-1] receptor agonists).

Target glucose levels may also be lower (fasting glucose levels of 70 to 110 mg/dL [4 to 6 mmol/L] and postprandial glucose levels 70 to 140 mg/dL [4 to 8 mmol/L]) in an effort to reduce complication risk; in addition, there is a lower risk of hypoglycemia in most children with type 2 diabetes (13).

Monogenic diabetes management

Management of monogenic diabetes is individualized and depends on subtype.

The glucokinase subtype generally does not require treatment because children are not at risk of long-term complications.

Most patients with the hepatic nuclear factor 4-alpha and hepatic nuclear factor 1-alpha subtypes are sensitive to sulfonylureas, but some ultimately require insulin. Other oral hypoglycemics such as metformin are typically not effective.

Screening for complications of diabetes

DKA is common among patients with type 1 diabetes; it develops in about 1 to 10% of patients each year, usually because they have not taken their insulin. Other risk factors for DKA include prior episodes of DKA, difficult social circumstances, depression or other psychiatric disturbances, and improper management of insulin needs during intercurrent illness. Interrupted delivery of insulin in children using an insulin pump (because of a kinked or dislodged catheter, poor insulin absorption due to infusion site inflammation, or pump malfunction) can also lead to rapid progression to DKA. Clinicians can help minimize the effects of risk factors by providing education, counseling, and support.

Mental health issues are very common among children with diabetes and their families. Up to half of children develop depression, anxiety, or other psychological issues. Eating disorders are a serious problem in adolescents, who sometimes also skip insulin doses in an effort to control weight. Psychological issues can also result in poor glycemic control by affecting children's ability to adhere to their dietary and/or medication regimens. Social workers and mental health professionals (as part of a multidisciplinary team) can help identify and alleviate psychosocial causes of poor glycemic control.

Vascular complications rarely are clinically evident in childhood. However, early pathologic changes and functional abnormalities may be present a few years after disease onset in type 1 diabetes; prolonged poor glycemic control is the greatest long-term risk factor for the development of vascular complications. Microvascular complications include diabetic nephropathy, retinopathy, and neuropathy. Microvascular complications are more common among children with type 2 diabetes than type 1 diabetes and in type 2 diabetes may be present at diagnosis or earlier in the disease course. Neuropathy is more common among children who have had diabetes for a long duration (≥ 5 years) with poor control (glycosylated hemoglobin [HbA1C] > 10%). Macrovascular complications include coronary artery disease, peripheral vascular disease, and stroke.

Patients are screened regularly for complications depending on the type of diabetes (see table Screening Children for Complications of Diabetes and Associated Disorders). If complications are detected, subsequent testing is done more frequently.

TABLE

Screening Children for Complications of Diabetes and Associated Disorders

Complication	Begin Screening	Screening Frequency	Method
Type 1 diabetes			
Hypertension	Upon diagnosis	Every visit	Blood pressure

Complication	Begin Screening	Screening Frequency	Method
			measurement
Dyslipidemia	Soon after diagnosis (once diabetes stabilized), age recommendations vary: • ADA: ≥ 10 years old • ISPAD: ≥ 11 years	ADA: If LDL ≤ 100 mg/dL (2.59 mmol/L), repeat at 9 to 11 years; then, if < 100 mg/dL, repeat every 3 years If abnormal, optimize glycemia control and nutritional therapy. For children > 10 years, if after 6 months LDL > 160 mg/dL (4.14 mmol/L) or > 130 mg/dL (3.37 mmol/L) with cardiovascular risk factors, initiate statin therapy. ISPAD: Every 3 years if results are normal	LDL, HDL, and triglyceride levels, nonfasting acceptable initially
Kidney disease	After 2 to 5 years of diabetes, start at puberty or at age 10 to 11 years (whichever is earlier)	If normal, annually ADA: If abnormal, repeat with confirmation in 2 of 3 samples over 6 months	Urine albumin:creatinine ratio
Neuropathy	After 2 to 5 years of diabetes, start at puberty or at age 10 to 11 years (whichever is earlier)	At regular visits, at least annually	Foot examination with foot pulses, pinprick, 10-g monofilament sensation tests,

Complication	Begin Screening	Screening Frequency	Method
			vibration, and ankle reflexes
Retinopathy	After 2 to 5 years of diabetes, start at puberty or at age 10 to 11 years (whichever is earlier)	2 to 4 years	Dilated examination by an ophthalmologist or other trained, experienced clinician
Celiac disease	Upon diagnosis	2 years *and* ADA: 5 years after diagnosis ISPAD: every 2–5 years	Celiac antibodies (tissue transglutaminase IgA if total IgA is normal; tissue transglutaminase IgG and anti-gliadin antibodies if total IgA is low)
Thyroid disease	Upon diagnosis	1 to 2 years if thyroid antibodies are negative, sooner if symptoms of hypothyroidism develop or thyroid antibodies are positive	At diagnosis: TSH and free T4 levels, thyroid antibodies (antithyroglobulin and antithyroid peroxidase) Serial monitoring: TSH with reflex to free T4 (if TSH is abnormal)
Psychosocial issues	Upon diagnosis	At least annually	Evaluation for diabetes distress, depression, disordered eating
Type 2 diabetes			
Hypertension	Upon diagnosis	Every visit	Blood pressure measurement
Dyslipidemia	Upon diagnosis	1 year	Same as type 1

Complication	Begin Screening	Screening Frequency	Method
Kidney disease (nephropathy)	Upon diagnosis	1 year	Same as type 1
Neuropathy	Upon diagnosis	At regular visits, at least annually	Same as type 1
Retinopathy	Upon diagnosis	1 year	Same as type 1
Psychosocial issues	Upon diagnosis	At least annually	Same as type 1
Obstructive sleep apnea	Not specified	ADA: At each visit	By history, with referral for sleep evaluation if indicated
Metabolic dysfunction-associated liver disease	At diagnosis	ADA: 1 year	ALT, AST

Data from American Diabetes Association Professional Practice Committee. 14. Children and Adolescents: Standards of Care in Diabetes-2025. Diabetes Care. 2025;48(1 Suppl 1):S283-S305. doi:10.2337/dc25-S014 and from Limbert C, Tinti D, Malik F, et al. ISPAD Clinical Practice Consensus Guidelines 2022: The delivery of ambulatory diabetes care to children and adolescents with diabetes. *Pediatr Diabetes.* 23(8):1243-1269, 2022. doi: 10.1111/pedi.13417.

ADA = American Diabetes Association; ALT = alanine aminotransferase; AST = aspartate aminotransferase; HDL = high-density lipoprotein; LDL = low-density lipoprotein; ISPAD = International Society for Pediatric and Adolescent Diabetes; free T4 = free serum thyroxine; TSH = thyroid-stimulating hormone.

Complications detected on examination or screening are treated first with lifestyle interventions: increased exercise, dietary changes (particularly limiting saturated fat intake), and cessation of smoking (if applicable).

Children with microalbuminuria (albumin/creatinine ratio 30 to 300 mg/g) on repeat samples or with persistently elevated blood pressure readings (> 90th to 95th percentiles for age or ≥ 130/80 mm Hg for adolescents) who do not respond to lifestyle interventions typically require antihypertensive therapy, most commonly using an angiotensin-converting enzyme inhibitor.

For children with dyslipidemia, if low-density lipoprotein (LDL) cholesterol remains > 160 mg/dL (4.14 mmol/L) or > 130 mg/dL (3.37 mmol/L) and 1 or more cardiovascular risk factors remain despite lifestyle

interventions, statins should be considered in children > 10 years, although long-term safety is not established. Target LDL is < 100 mg/dL (2.59 mmol/L).

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Screening for Diabetes in Children and Adolescents

Asymptomatic children ≤ 18 years of age who are at risk should be screened for type 2 diabetes or prediabetes by measuring HbA1C. This test should first be done at age 10 years or at onset of puberty, if puberty occurred at a younger age, and should be repeated at a minimum every 3 years. Annual screening may be necessary in a child whose BMI has increased or whose cardiometabolic profile has worsened, who has strong family history of type 2 diabetes, or who has evidence of prediabetes (1).

Children at risk include those with overweight (body mass index > 85th percentile for age and sex, or weight for height > 85th percentile) and who have any 2 of the following:

- Family history of type 2 diabetes in a first- or second-degree relative
- Native American, Black, Hispanic, Asian American, and Pacific Islander heritage
- Signs of insulin resistance or conditions associated with insulin resistance (eg, acanthosis nigricans, hypertension, dyslipidemia, polycystic ovary syndrome, or small-for-gestational-age birth weight)
- Maternal history of diabetes or gestational diabetes

Screening reference

1. Shah AS, Zeitler PS, Wong J, et al: ISPAD Clinical Practice Consensus Guidelines 2022: Type 2 diabetes in children and adolescents. *Pediatr Diabetes* 23(7):872-902, 2022. doi: 10.1111/pedi.13409

Key Points

- **Type 1 diabetes is caused by an autoimmune attack on pancreatic beta-cells, causing complete lack of insulin; it accounts for two thirds of new cases in children and can occur at any age.**
- **Type 2 diabetes is caused by insulin resistance and relative insulin deficiency due to a complex interaction among many genetic and environmental factors (particularly obesity); it is increasing in frequency in children and occurs after puberty.**
- **Most children have symptomatic hyperglycemia without acidosis, with several days to weeks of urinary frequency, polydipsia, and polyuria; children with type 1 diabetes and rarely type 2 diabetes may present with diabetic ketoacidosis.**
- **Screen asymptomatic, at-risk children for type 2 diabetes or prediabetes.**
- **All children with type 1 diabetes require insulin treatment; intensive glycemic control helps prevent long-term complications but increases risk of hypoglycemic episodes.**
- **Advances in diabetes technology, such as continuous glucose monitoring systems, are aimed at improving glycemic control while reducing hypoglycemic episodes.**
- **Children with type 2 diabetes are initially treated with metformin and/or insulin; although most children requiring insulin at diagnosis can be successfully transitioned to**

metformin monotherapy, about half eventually require insulin treatment.

- GLP-1 agonists can be used in combination with metformin to improve glycemic control.
- Mental health issues are common among children with diabetes and can be associated with poor glycemic control.
- Insulin doses are adjusted based on frequent glucose monitoring and anticipated carbohydrate intake and activity levels.
- Children are at risk of microvascular and macrovascular complications of diabetes, which must be evaluated by regular screening tests.

More Information

The following English-language resources may be useful. Please note that THE MANUAL is not responsible for the content of these resources.

American Diabetes Association: 14. Children and Adolescents: Standards of Care in Diabetes—2023

International Society for Pediatric and Adolescent Diabetes (ISPAD): Clinical practice consensus guidelines for diabetes in children and adolescents (2022)

Type 1 Diabetes TrialNet: Pathway to Prevention: Study Details: A resource providing information about how to receive screening and enroll in prevention studies

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